Pet bird viral diseases

陳慧文 DVM PhD

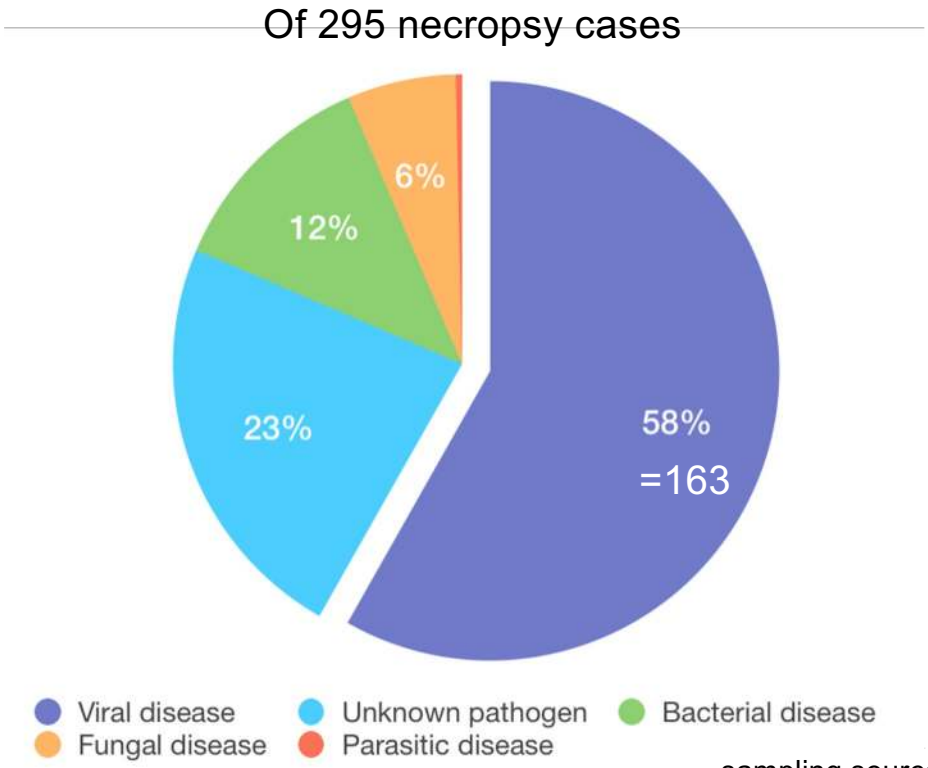
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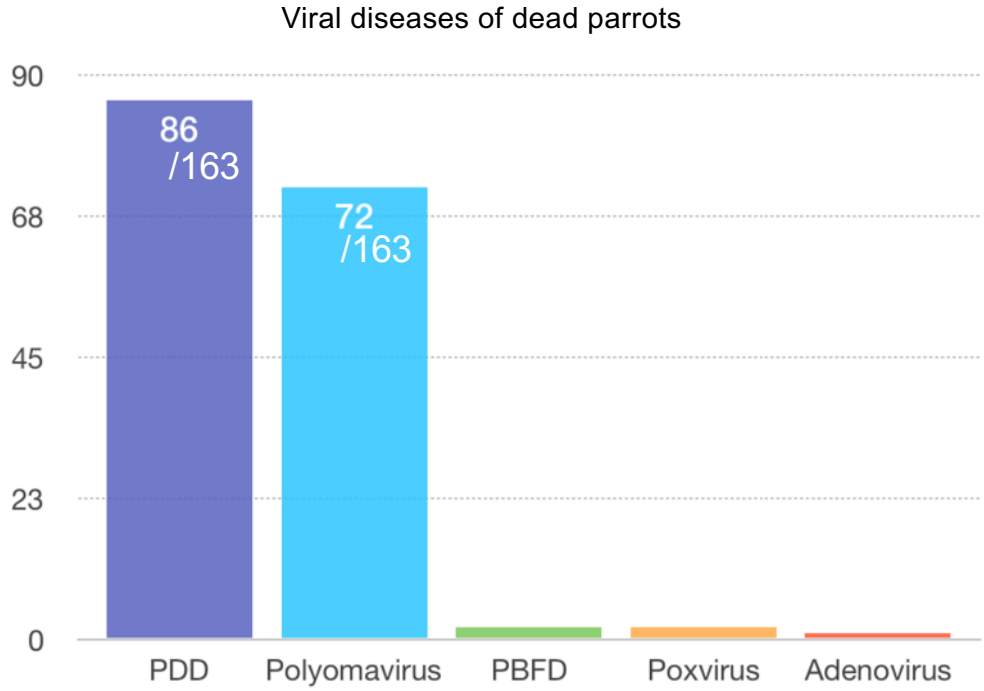
寵物鳥/鸚鵡常見病毒感染症

- 玻那病毒感染症 (PDD)
- 多瘤病毒感染症 (Avian polyomavirus)
- 環狀病毒感染症 (PBFD)
- 痘病毒感染症 (Avian pox)

Common viral diseases of pet birds

2014-2017 高雄市動保處統計資料 [以死亡剖檢動物為樣本]





sampling source: dead parrots from breeder, bird farms, bird shops, private owned birds

Parrot bornavirus (PaBV)
玻那病毒感染症



Proventricular dilatation Disease (PDD)

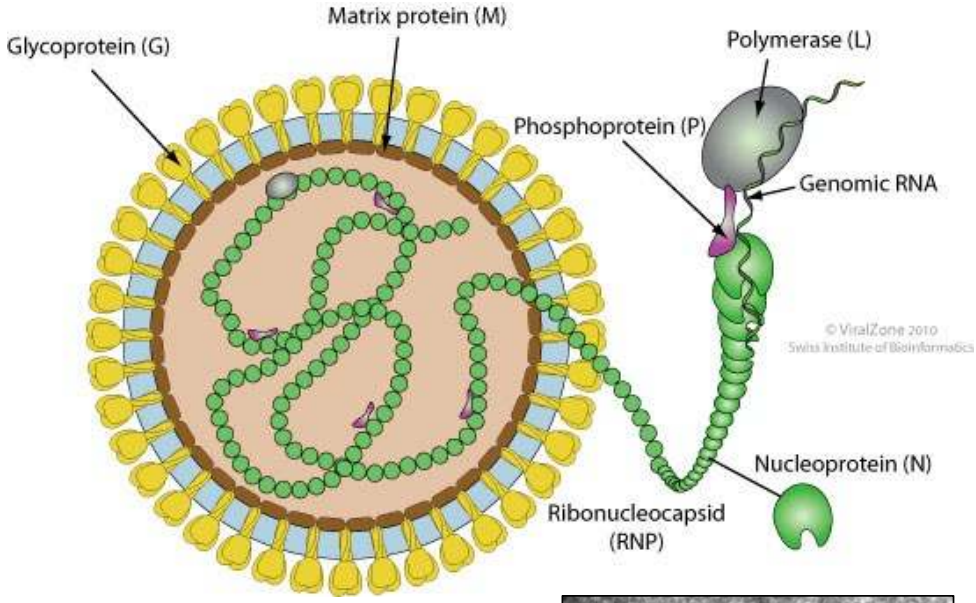
- Previously know as “Macaw Wasting Disease” in 1970s.
- PDD is recognized by the International Society of Aviculture (IAS) as an extremely dangerous disease.
- Fatal and neuropathic.
- The etiological agent was identified in 2008 as the *Bornavirus*.

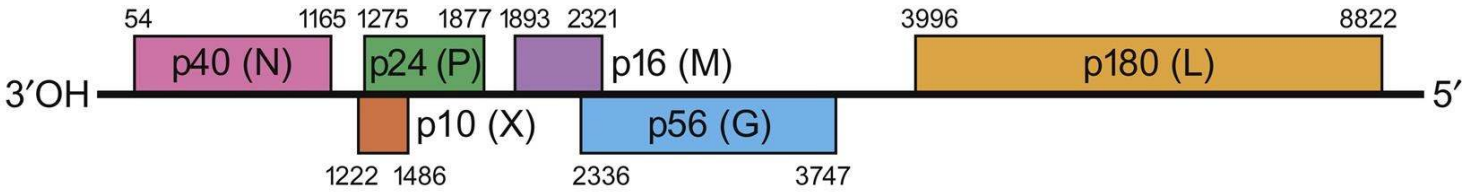
(Honkavuori et al., 2008; Kistler et al., 2008)

Avian bornavirus (ABV)

- Bornaviruses of avian origin were first discovered in psittacines in 2008 and subsequently also identified in waterfowls and songbirds.

- The virus is highly neurotropic and infects the central and peripheral nervous systems = avian ganglioneuritis (AG)







Affected bird species

- PDD-like features have been reported in >80 avian species.
- African grey parrots (*Psittacus erithacus*), macaws (*Ara* species), Amazon parrots (*Amazona* species), and cockatoos (*Cacatua* species) are the most common psittacine species affected.







- The disease is minimally represented in Quaker parrots (*Myiopsitta monachus*) and lovebird species (*Agapornis*).




Transmission

- Fecal-oral, fecal-oronasal, urinary transmission
- Vertical transmission
  - 2 out 30 embryos (brain, proventriculus) from PaBV-infected flocks were detected positive. (Lierz et al., 2011)
  - 4 out of 4 embryos (brain, liver, eyes) were detected positive for viral RNA. (Kerski et al., 2012)
- Persistent infection
  - 29 out of 31 (94%) macaws which survived an outbreak showed the presence of virus. (Kloet et al., 2011)
  - IV or IC infection induced persistent infection up to 33 weeks. (Piepenbring et al., 2012)

Clinical signs

- Neurological signs 66%, including depression, convulsions, ataxia, blindness, incoordination
- Undigested food in stools 22%
- Regurgitation 19%
- Proventricular dilatation 13%
- Feather plucking 9%
- Sudden death 9%

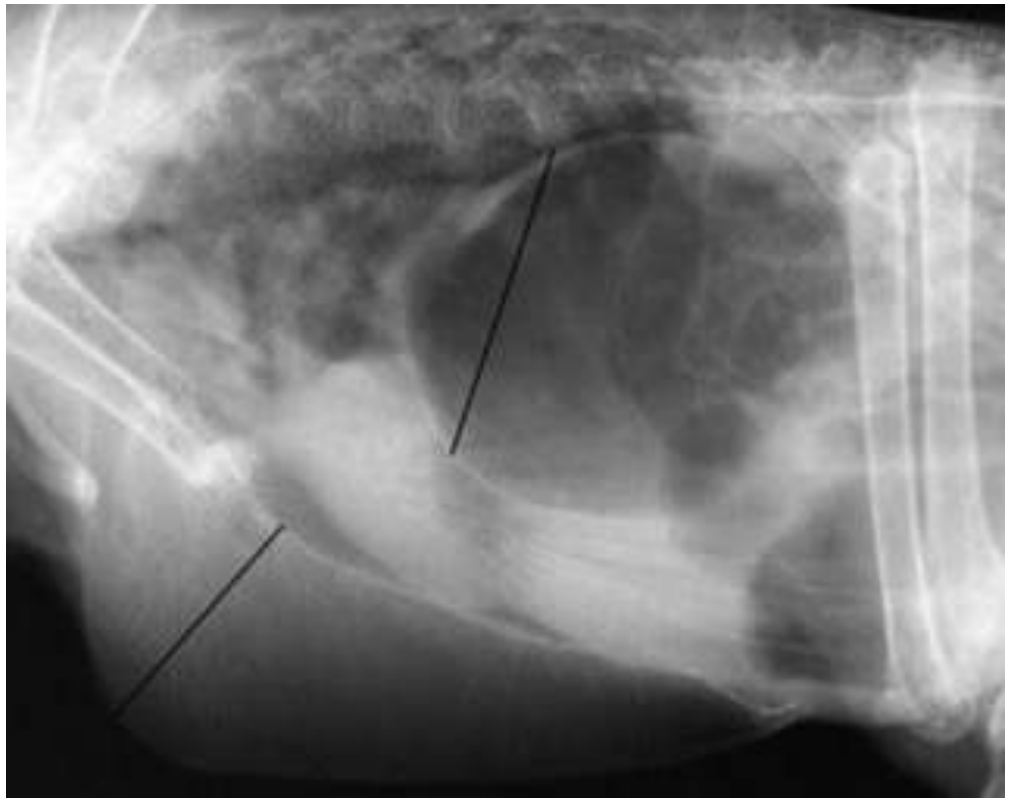




(Philadelpho et al., 2014)

X-ray evaluation

- Proventriculus: keel ratio (normal : <0.48)



proventriculus : keel ratio =1.37

(Dennison et al., 2009)

Gross findings



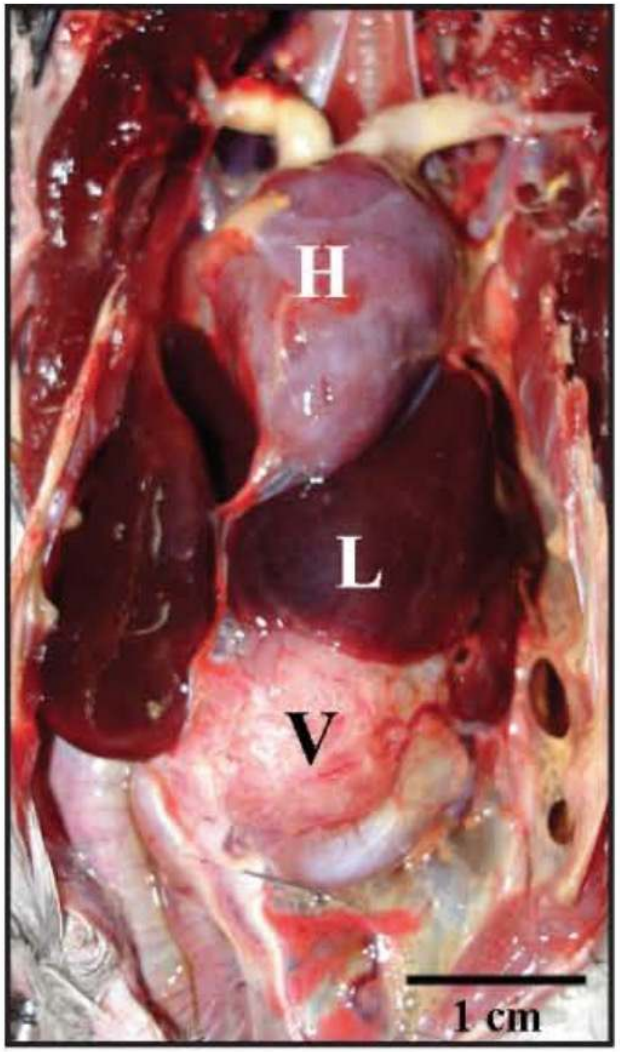
Amazona rhodocorytha in poor body condition



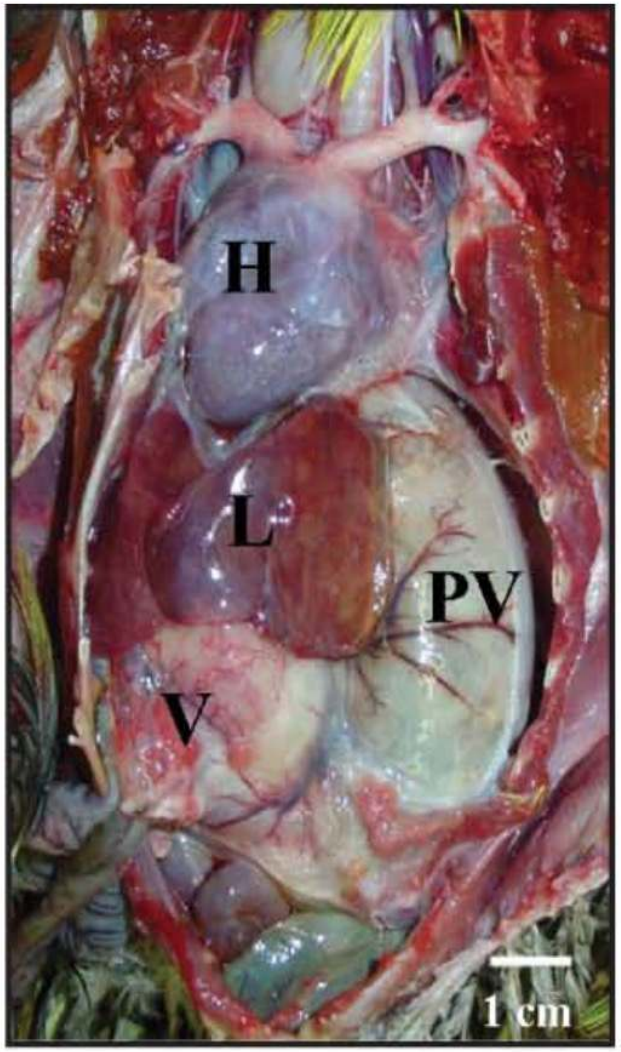
Proventriculus and ventriculus markedly distended with undigested food

(Donatti et al., 2014)

Normal

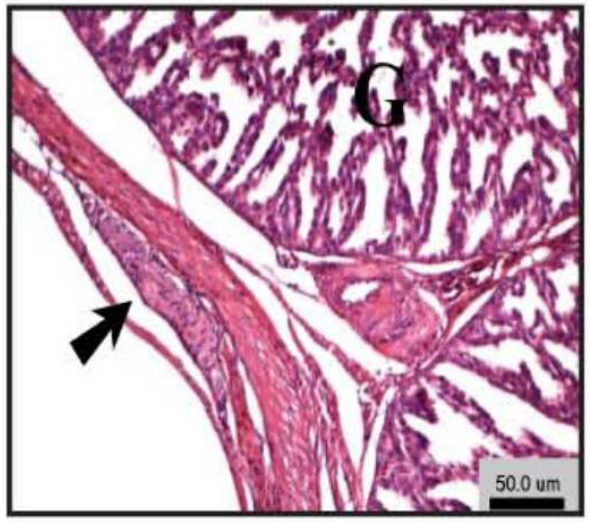


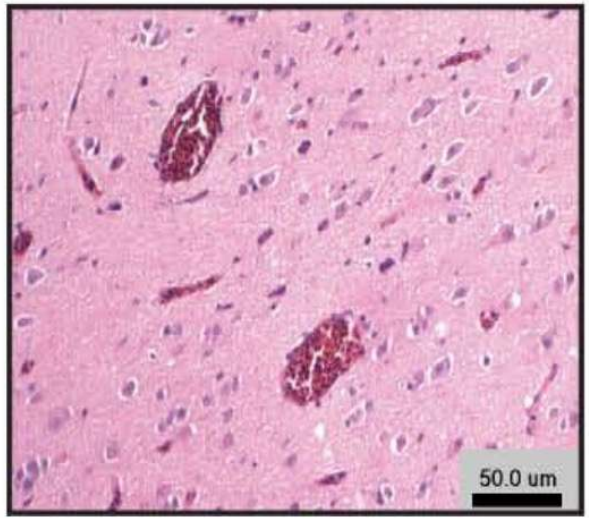
Diseased African grey parrot



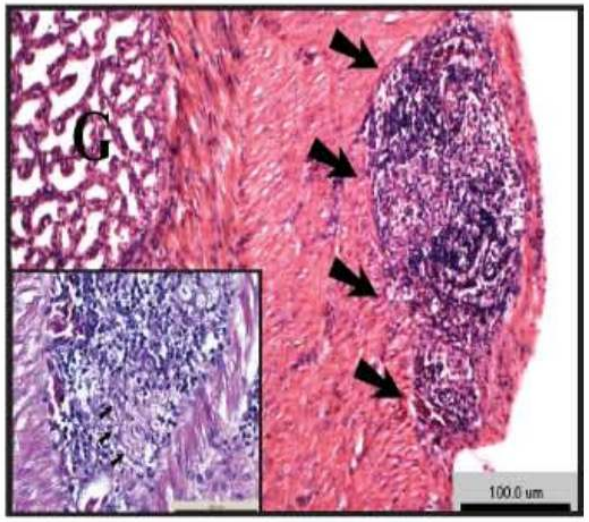
(Kistler et al., 2008)

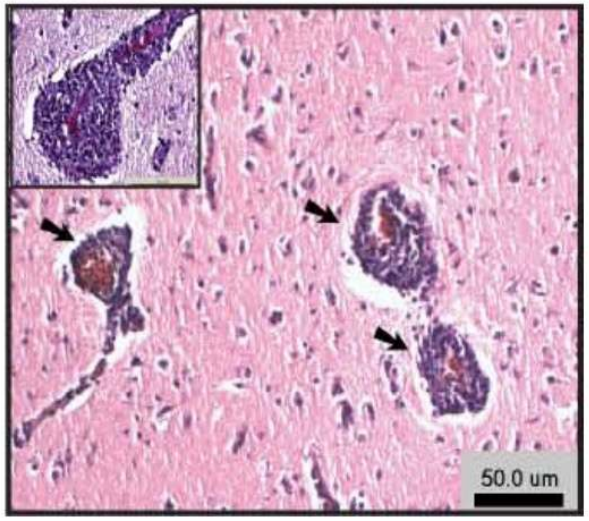
Normal





Diseased African grey parrot





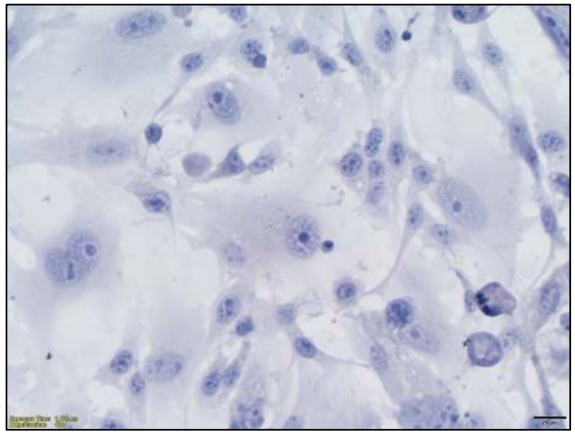
Diagnosis: virus detection

- Positive rate for RT-PCR:
  - Cloacal swabs 53%
  - Feather 31%
  - Feces 15%
  - Blood 0%

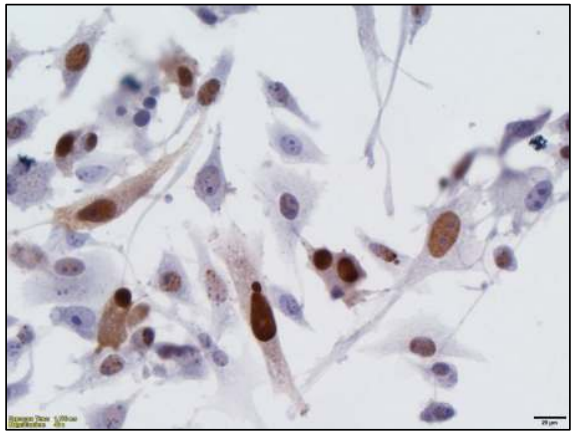
(Philadelpho et al., 2014)

Virus isolation using QM7 cells

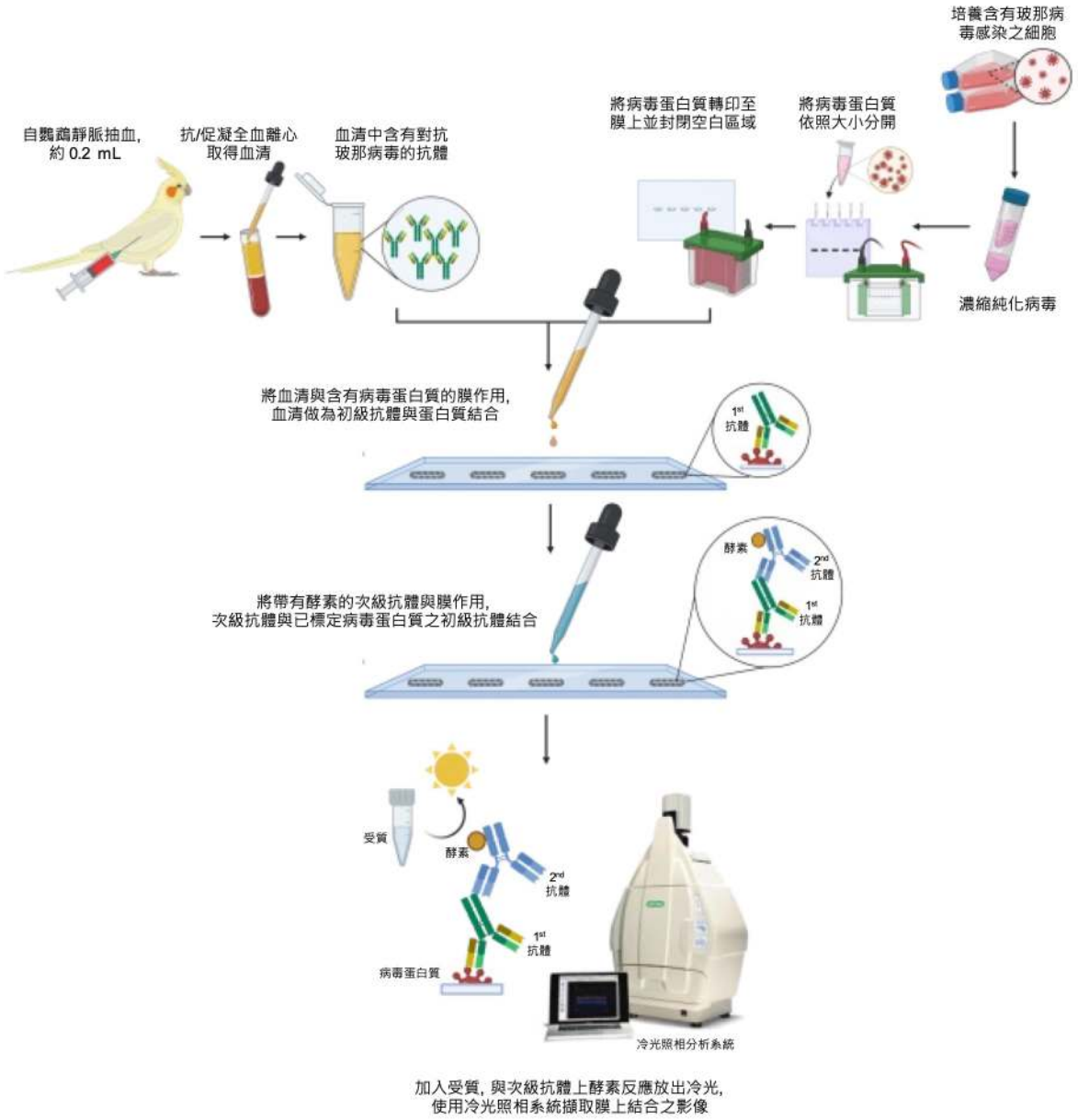
Un-infected QM7

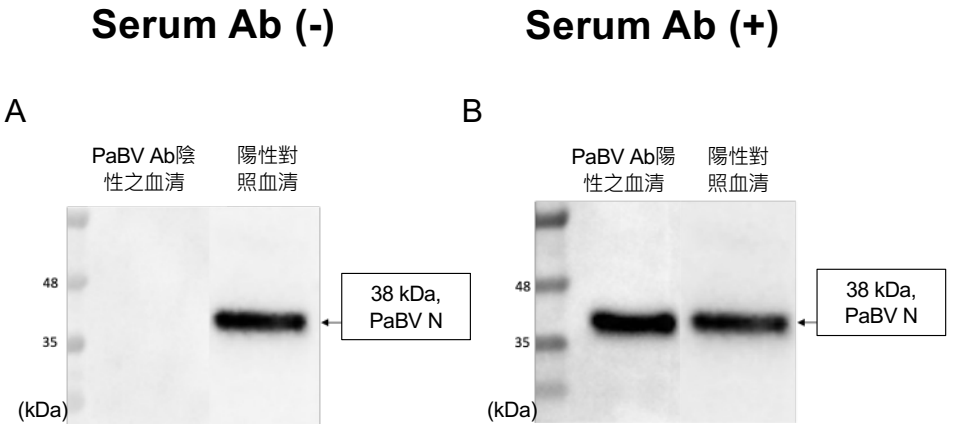


PaBV infected QM7



Diagnosis: Ab detection





(陳慧文實驗室)

Medications

- Celecoxib has been used successfully in a number of avian patients as well as the use of robenacoxib by injection.

- For those avian patients with alteration of normal proximal GI tract function, injectable metoclopramide is preferred.

Avian Ganglioneuritis in Clinical Practice

Vet Clin Exot Anim 21 (2018) 33–67

Giacomo Rossi, DVM, PhD, DECZM (Wildlife Population Health)[a],
Robert D. Dahlhausen, DVM, MS[b], Livio Galosi, DVM[a],
Susan E. Orosz, PhD, DVM, DABVP (Avian), DECZM (Avian)[c,*]

Prevention and control

- No vaccines available
  - Antibody does not seem to be able to clear the virus.
  - Studies on the subunit or attenuated vaccines are ongoing.
  - Cross-genotype protection?
- No antivirals available
  - drug under investigation: Ribavirin, Favipiravir

(Mirosseini et al., 2011; Obert et al., 2016; Runge et al., 2017; Takunaga et al., 2017)

Treatment

- Quarantine (foreign → domestic)
- Disinfection

- Supportive care with fluids, antibiotics, and feeding small amounts of food at each meal
- With patience, perseverance, prolonged therapy, reduced stress, and attention to correction of secondary problems, quality and longevity of life in many PaBV clinically affected birds can be significantly improved.

Summary

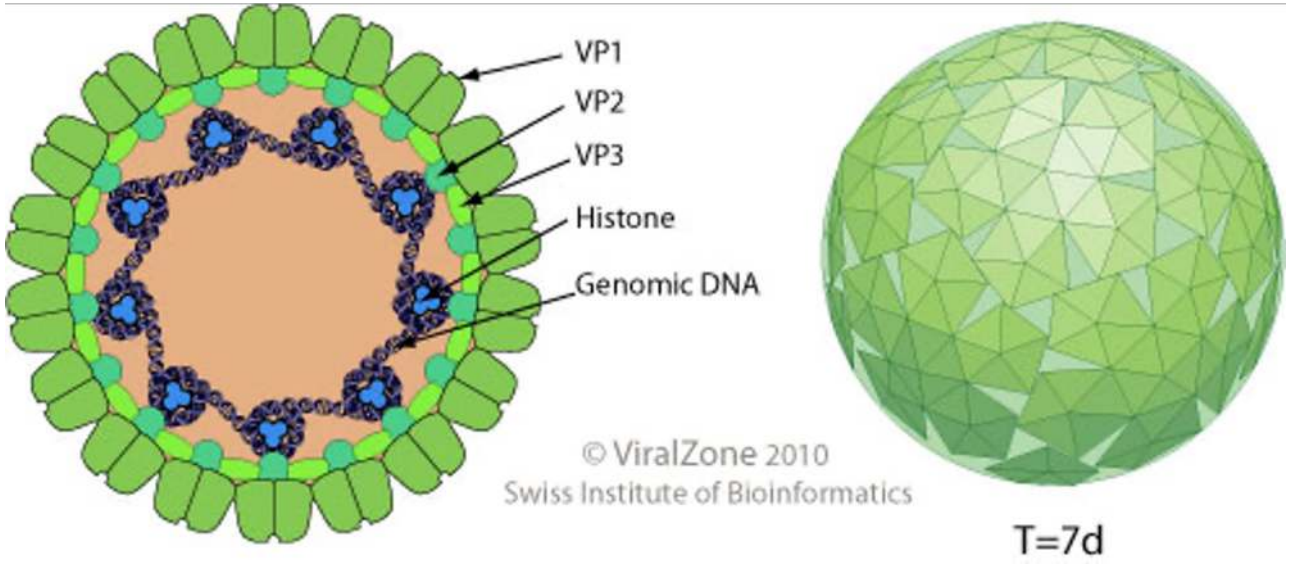
- 感染多種品種之鸚鵡，如灰鸚鵡、鳳頭鸚鵡、亞馬遜、金剛等
- 病毒潛伏性感染、從無症狀到高死亡率變異大
- 各臟器之神經系統受影響，產生行為異常、消化道遲滯、心律不整、眼盲、昏厥、猝死等
- 依其病毒基因型可分為8型，其抗原性差異仍未知
- 台灣盛行率高、藥物與疫苗待研發

Avian Polyomaviruses (APyVs)
多瘤病毒感染症



Structure of Polyomavirus

- Circular dsDNA (~5 kbp)
- Non-enveloped
- Spherical in outline
- Replicate in the nucleus



https://viralzone.expasy.org/148?outline=all_by_species

Characteristics of avian polyomaviruses

- APyV was first discovered in budgerigars in 1981 (Budgerigar fledgling disease)
- Budgerigars虎皮, lovebirds愛情鳥 and cockatiels玄鳳 are highly susceptible whereas cockatoo鳳頭 is resistant
- The infection is more fatal in young birds of less than 16 weeks age

Breed	Age of peak susceptibility	Shedding period
Budgerigar	10–25 days old	up to six months
Conure	<6 weeks	4–8 weeks
Macaws and eclectus	<14 weeks old	6–14 weeks
Lovebird	up to 1 year	-

Characteristics of avian polyomaviruses

- The nestling mortality rate is high and may approach 100%
- The degree of susceptibility, the tissue tropism, and the resulting disease appears to be dependent on the species and age
  - Larger psittacine birds (parrots) ⇒ targets mononuclear phagocytic cells
  - Passerines ⇒ targets endothelial cells

APV-positive rate in different groups in Taiwan

- The APyVs-positive rate: about 15% in Taiwan
  - ✓(15.2% (22/165) from clinical cases in 2002-2005, 16.56% (27/163) from bird farms in 2010-2014)

- APV/PBFDV dual infection rate: about 10% in Taiwan
  - ✓(10.3% (17/165) from clinical cases in 2002-2005,11.04% (18/163) from bird farms in 2010-2014)

(Hsu, C.M et al ,2006; Duangsuda Thongchan et al, 2015)

Polyomavirus- transmission



- Horizontal transmission:
  - ✓Feces, secretion, contaminated cages, egg incubator

- Vertical transmission
  - ✓Has been reported in budgerigars and other species suspected

https://exoticbirdhospital.com/portfolio/budgerigar/

Polyomavirus - clinical signs

- May be asymptomatic
- Feather dysplasia of primary wing feathers and tail feathers-> 'French molt'
- 'Runners' or 'Creepers'
- Crop stasis
- Abdominal distension
- Neurological signs

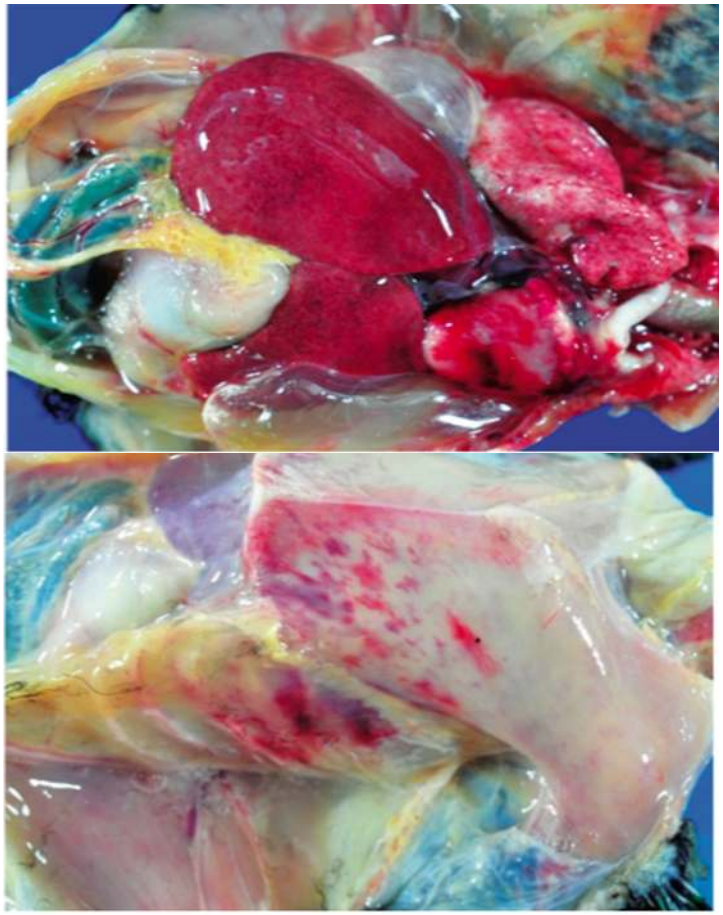




http://www.birds-online.de/gesundheit/gesgefieder/franzmauser_en.htm

Polyomavirus - gross lesions

- Cardiomegaly, hepatomegaly, splenomegaly
- Petechial hemorrhages
- Subcutaneous hemorrhages
- Hydropericardium, ascites
- Muscles may appear an orange color (pale)
- Bleeding from feather follicles

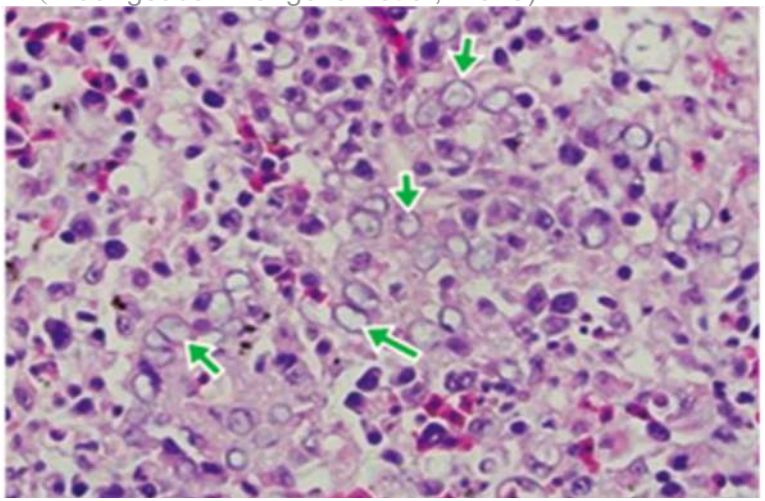


(Duangsuda Thongchan et al, 2015)

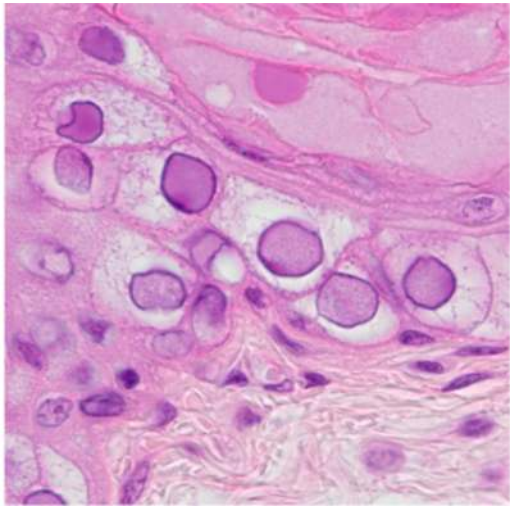
Polyomavirus - histological lesions

- Liver, spleen, kidney necrosis
- Basophilic intranuclear inclusions in the spleen, mesangial cells in the kidneys and in Kupffer cells
- In budgerigars, often found in the feather follicles and growing feathers

(Duangsuda Thongchan et al, 2015)



(CASE III: S394/10 (AFIP 3164881), WSC 2010-2011)



Polyomavirus - diagnosis

- PCR (blood, cloacal swabs from live animals and tissues from spleen, liver and kidney)
- Immunofluorescent antibody staining
- Electron microscopy

Polyomavirus - diagnosis - PCR

- After initial infection there is a viremia → blood PCR (+)
- After a short period, the virus can be detected in the cloaca PCR (+)
- When stop shedding→ the blood PCR (-)
- After 6-8 weeks the cloacal shedding stops→ cloaca PCR (-)

blood PCR/cloaca PCR	Result
+/-	early stages of infection, or intermittent shedding
+/+	actively infected and shedding
-/+	bird is about to stop shedding
-/-	bird has cleared infection, no longer shedding

(Avian medicine and surgery in practice, companion and aviary birds, 2016)

Polyomavirus - treatment and prevention

- Supportive care
- Vaccine (Psittimune® APV)
  - Primary vaccination is done at 5 weeks and booster after 2–3 weeks
- Disinfection: synthetic phenol, sodium hypochlorite, stabilized chlorine dioxide



https://creativesciencellc.com/psittimune-apv/

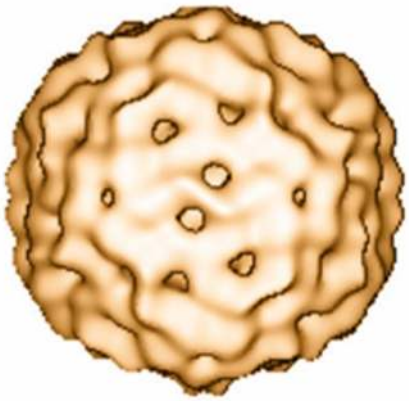
Psittacine beak and feather disease (PBFD)/Circovirus infection

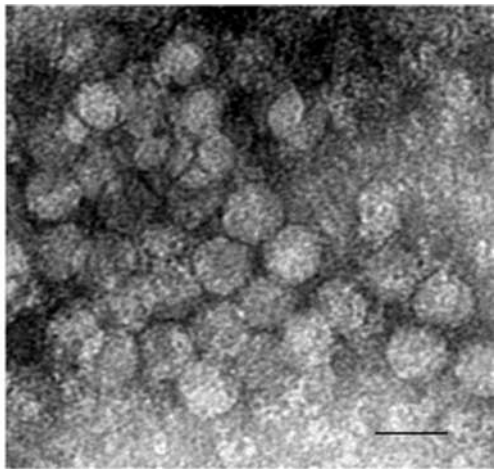
鸚鵡喙羽病/環狀病毒感染症



Structure of circovirus

- Circular ssDNA (~ 2 kb)
- Very small (approximately 20-25 nm in diameter)
- Non-enveloped
- Spherical in outline
- Highly stable in the environment





https://talk.ictvonline.org/ictv-reports/ictv_online_report/ssdna-viruses/w/circoviridae/659/genus-circovirus

Circovirus of pet birds

- Cockatoos, African gray parrots, budgerigars, macaws, Amazon parrots
- Wild birds can be carriers/reservoirs
- Young birds during first feather formation
- High morbidity but low mortality
- Incubation period can be very long



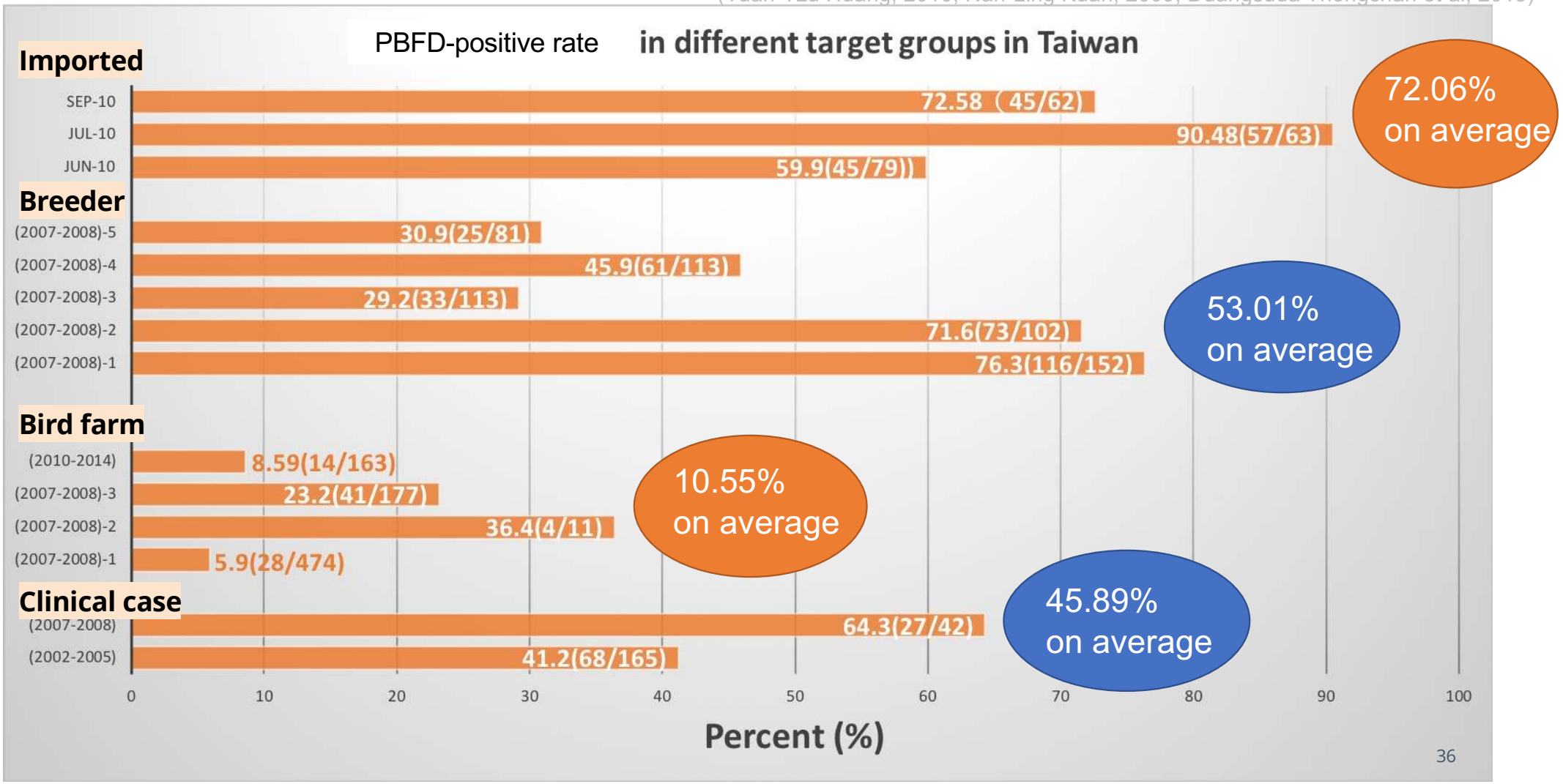




https://www.singing-wings-aviary.com/african-grey-parrot.htm

PBFD-positive rate in Taiwan

(Yuan-Tzu Huang, 2010; Nan-Ling Kuan, 2009, Duangsuda Thongchan et al, 2015)



PBFD - transmission



- Horizontal transmission
  - feces and feather dander
  - regurgitated crop contents
  - non-psittacine birds may play as reservoirs of infection

- Vertical transmission:
  - infected hens to embryonated eggs has been reported

http://www.kcbbs.gen.nz/lori/ar/behaviour.html

Pathogenesis and Pathology

- Requirement for actively dividing cells to facilitate replication of their DNA

- Basal epithelial layer of the feather follicles, beak and claw, as well as in lymphoid tissues (immune suppression) and intestinal epithelium



https://veteriankey.com/feather-follicle-extirpation/



PBFD - clinical signs

- Peracute
  - Diarrhea-causing enteritis or pneumonia
  - Die with acute hepatic necrosis

- Acute
  - Die within days to weeks of the first signs of feather abnormalities

PBFD - clinical signs



- Chronic
  - Weight loss, lethargy, anemia, diarrhea
  - Shedding of developing feathers
  - Contour, tail and down feathers are lost symmetrically and they are replaced with dystrophic feathers
  - Lose primary and secondary wing feathers
  - Feather color change





(Yvonne R.A. van Zeeland et al, 2014, Shane Raidal, 2016)

PBFD - clinical signs

- Chronic
  - Beak abnormalities (shiny, overgrown, broken, exhibiting delaminations, palatine necrosis)
  - Immunosuppression
  - Secondary infections
- Hematology
  - Pancytopenia
  - nonregenerative anemia



https://cockatoo-info.com/health/pbfd/

PBFD - gross lesions

- Feather lesions, beak lesions or both
- Abnormal pin feathers (constricted, clubbed or stunted)
- Abnormal mature feathers (retention of sheaths, blood in shaft, fracture of rachis)



FIGURE 2-10 A dystrophic feather, showing blood within the calamus and annular constrictions of the calamus.

(Shane Raidal, 2016)

近端羽軸發育不良、彎曲



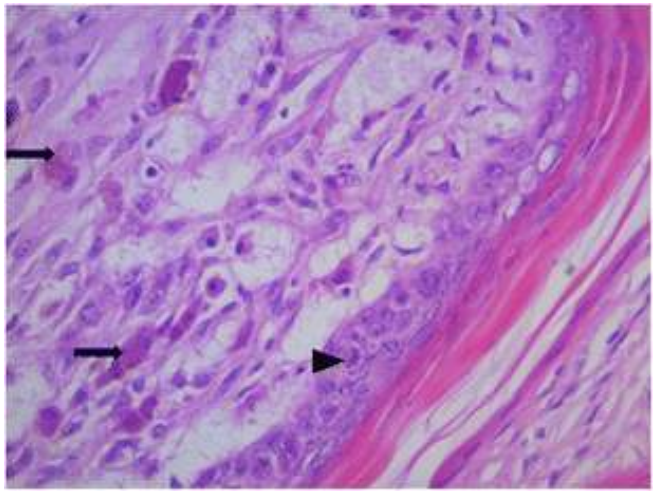
Figure 20. Annular constriction and bending of the feathers in a bird with PBFD.

PBFD - histological lesions

- Basophilic intracytoplasmic inclusion bodies
  - Follicular epithelium, macrophages and the epithelium of the cloacal bursa

- Lymphoid depletion (bursa, thymus)
- Multifocal necrosis of epidermal cells
- epidermal hyperplasia
- epidermal hyperkeratosis

Sa et al., 2014



Macrophages in the pulp show prominent cytoplasmic inclusion bodies (*arrows*).
Basal epithelial cells show marginated chromatin and nuclear inclusion bodies (*arrowhead*)

PBFD - diagnosis

- Signalment and clinical signs
- Basophilic intracytoplasmic inclusion bodies (follicular epithelium, macrophages and the epithelium of the cloacal bursa)
- PCR (blood, fresh feather pulp, feces)
- Electron microscopy
- In situ hybridization

PBFD - treatment and prevention

- Supportive care and antimicrobials to secondary infections
- Strict isolation or culling of infected birds
- Disinfection: 2% glutaraldehyde
- No commercial vaccine





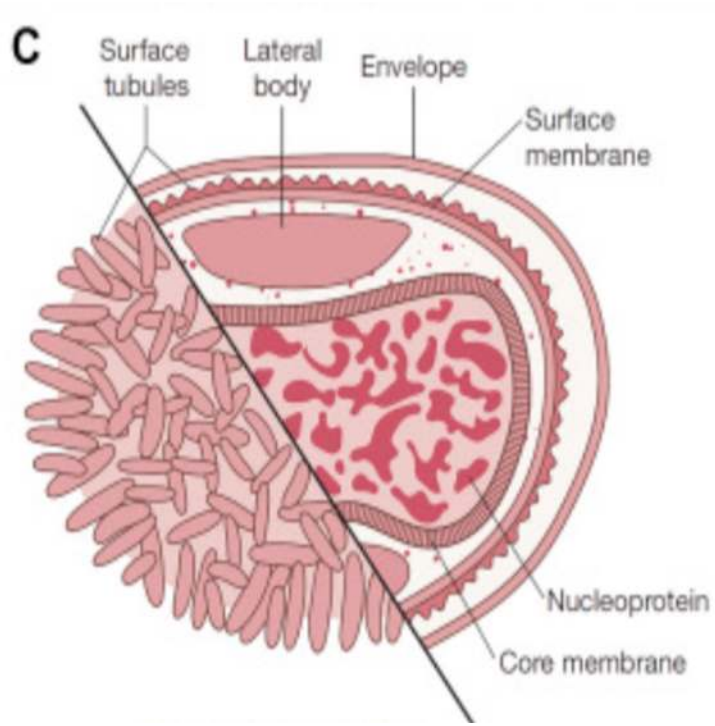
https://www.wikihow.com/Treat-Psittacine-Beak-and-Feather-Disease-in-Lories-and-Lorikeets

Avian Poxvirus
痘病毒感染症



Structure of Poxvirus

- Brick-shaped virion, 200 nm
- Enveloped
- Linear dsDNA (100-300 kbp)



https://www.sciencedirect.com/topics/neuroscience/poxviruses

Characteristics of Avipoxvirus

- Survive easily in the environment, especially in scabs





https://forum.birdsplanet.com/forum/birds/birds-general-discussion/28316-pox-virus-treatment-specially-for-love-birds

Avipoxvirus of pet birds

- Species-specific poxviruses
  - Psittacine poxvirus: Amazon parrots, macaws
  - Agapornic poxvirus: lovebirds
  - Canary poxvirus: canaries
- Infects other hosts, but causes the severest symptom in the natural host



G. Gonzalez-Hein et al. (2008) Case Report: An Avian Pox Outbreak in Captive Psittacine Birds in Chile

Poxvirus - species specific poxvirus

- Canary pox
  - Septicemic form and lung tumors
  - High mortality (over 70%)



https://www.memorangapp.com/flashcards/210071/Infectious+Diseases+53-54/

Poxvirus - transmission



- Virus enters through
  - Open wounds
  - Inhalation or digestion⇒ damaged mucosa

- Virus can be carried by bloodsucking insects (mosquitoes, mites) or aerosols



http://lookfordiagnosis.com/mesh_info.php?term=variole+aviaire&lang=4

http://foam-friends-of-albatross-on-midway.blogspot.com/2010_05_01_archive.html

Poxvirus - clinical signs



- 10-14 days after infection
- Cutaneous form (dry pox)
  - Skin surface lack of feathers (eyelid margins, face, feet): erythematous macules-papular-vesicular-pustulues-scab-scar
- Diphtheritic form (wet pox)
  - Eyes: blepharitis, ocular discharge, conjunctivitis
  - Nose: rhinitis
- Stress may lead to latent infection

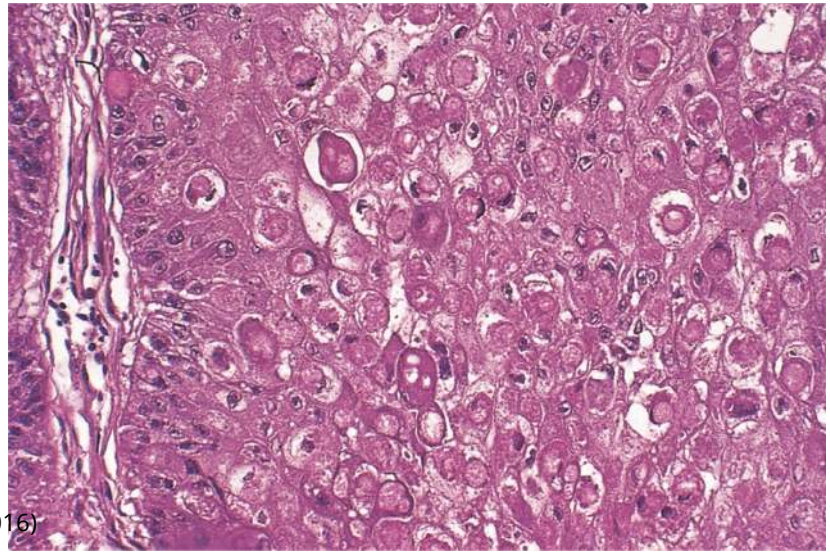


https://www.memorangapp.com/flashcards/210071/Infectious+Diseases+53-54/

https://www.youtube.com/watch?v=1NI2v74GhFQ

Poxvirus - histological lesions

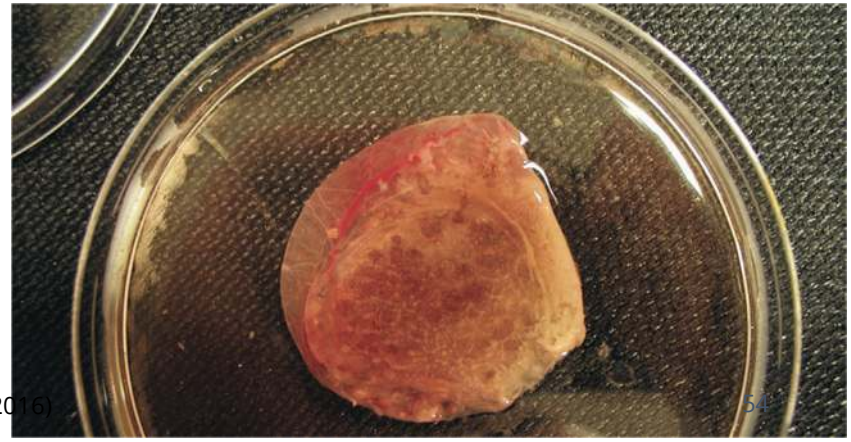
- Skin and mucosa cells: intracytoplasmic inclusion bodies (Bollinger bodies)
- Epithelial cells: vacuolation, ballooning degeneration
- Edema, vascular dilation, perivascular mononuclear and neutrophilic cell infiltration



Jaime Samour, Avian Medicine third edition (2016)

Poxvirus - diagnosis

- Gross lesions observation
- Histological characteristics
- PCR
- Electron microscopy - very big virions, easily to be detected
- Virus inoculation: pocks can be seen on chorioallantoic membrane of fertilized egg



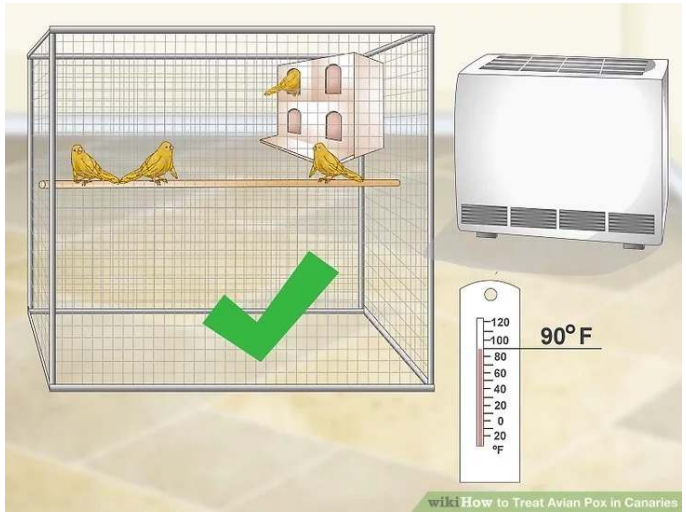
Jaime Samour, Avian Medicine third edition (2016)

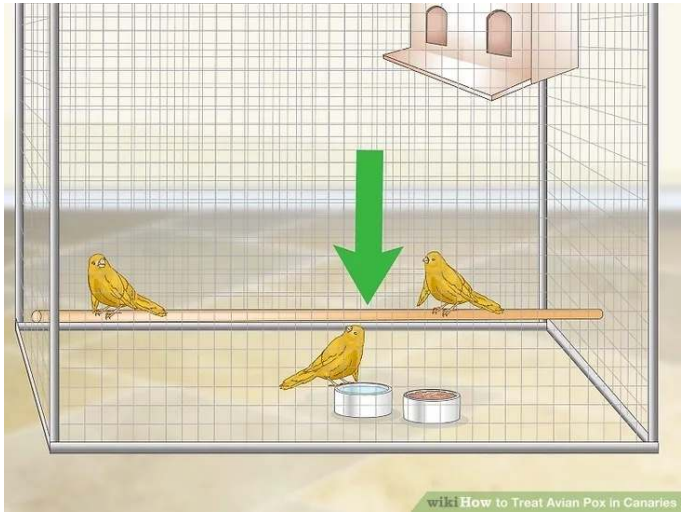
Poxvirus - treatment and prevention

- Supportive care
- Treat secondary infections with antibiotics
- Let the pox lesions heal naturally

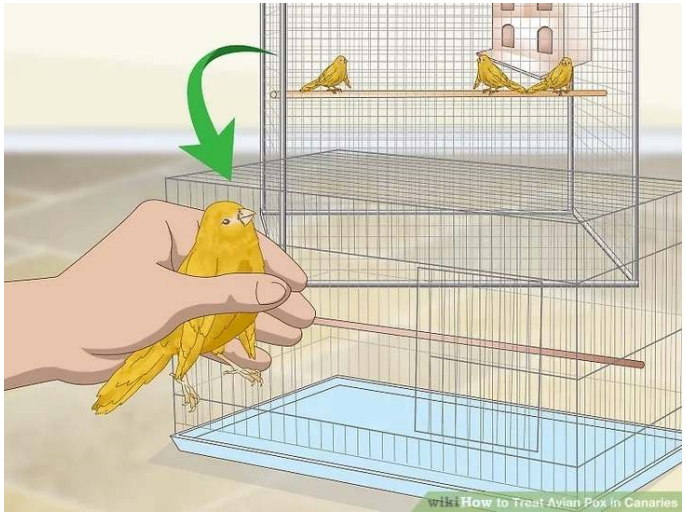
- Vaccination: canarypox vaccine (POXIMUNE®C) can be used in canaries
- Insecticide
- Hygiene: sodium hypochlorite







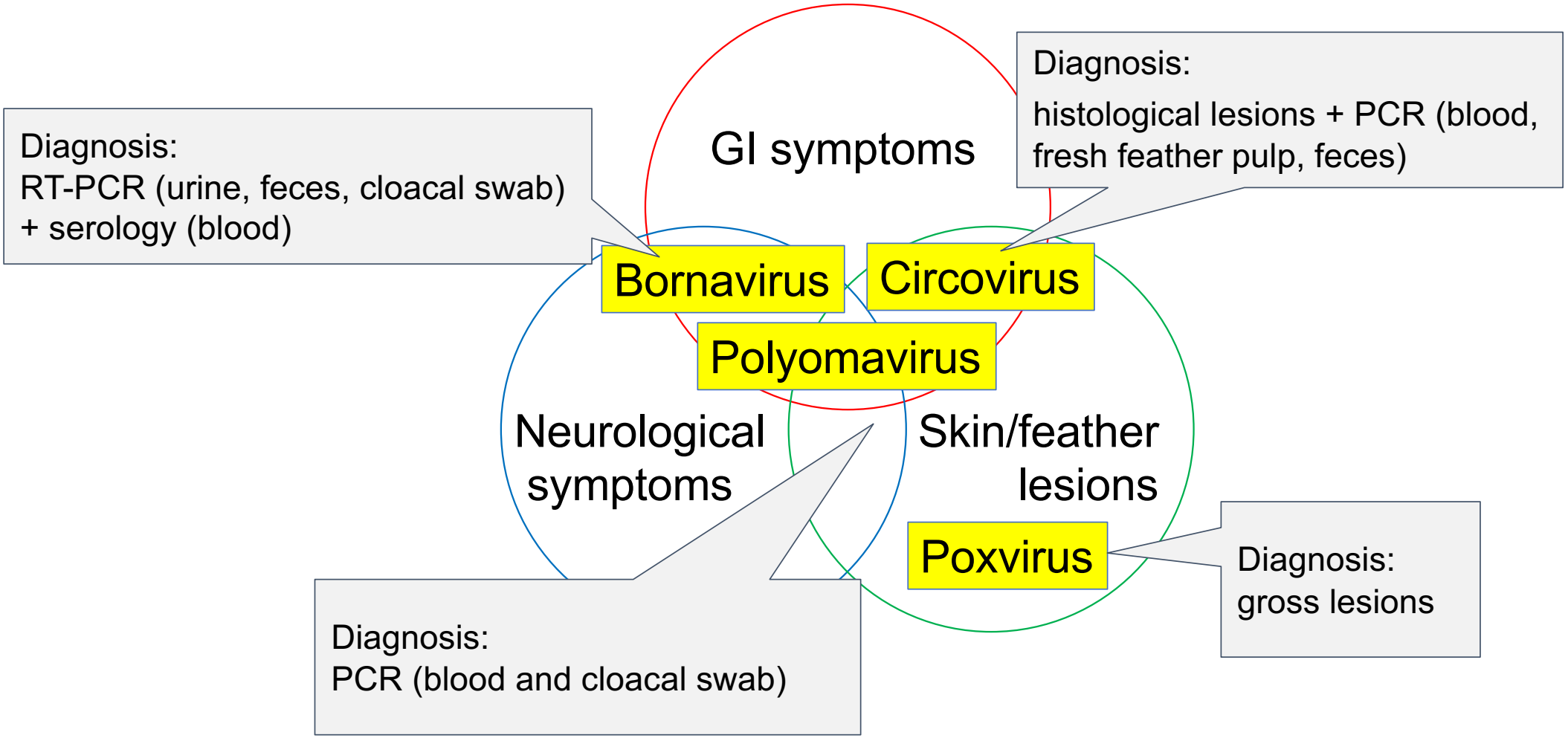
Providing Supportive Care







https://www.wikihow.com/Treat-Avian-Pox-in-Canaries


GI symptoms
Neurological symptoms
Skin/feather lesions
Bornavirus
Circovirus
Polyomavirus
Poxvirus
Diagnosis:
RT-PCR (urine, feces, cloacal swab) + serology (blood)
Diagnosis:
histological lesions + PCR (blood, fresh feather pulp, feces)
Diagnosis:
PCR (blood and cloacal swab)
Diagnosis:
gross lesions